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FERTILITY PRESERVATION

Fertility-related worry among emerging adult cancer survivors

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Abstract

Purpose Cancer survivors with a history of gonadotoxic treatment are at risk for future infertility and reproductive concerns, including worry about infertility. The purpose of this study was to describe factors associated with fertility-related worry among emerging adult survivors of childhood cancer.

Methods This chart review included patients aged 18.00–25.99 years and > 1 year from cancer treatment completion with a history of gonadotoxic treatment. Survivors were offered structured fertility-focused discussions at age ≥ 18 years, which assessed worry about future infertility. Data from this discussion (i.e., reported fertility-related worry (yes/no), sociodemographic, and clinical characteristics were abstracted from the medical record. Multivariable logistic regression with backwards elimination was used to calculate odds ratios (OR) and 95% confidence intervals (95%CI) for factors associated with fertility-related worry.

Results Survivors ($N = 249$) were a mean age of 19.1 ± 1.2 years at initial fertility discussion; 55.8% were male, 58.2% non-Hispanic White, and 27.3% were at high risk for future treatment-related infertility. Fertility-related worry was reported by 66.3% of survivors. Factors related to worry on multivariable analysis included female sex (OR: 2.64, 95%CI: 1.44–4.96, $p = .002$), solid tumor diagnosis (OR: 2.31, 95%CI: 1.15–4.71, $p = .019$), moderate and high risk of infertility (OR: 2.94, 95%CI: 1.23–7.64, $p = .02$; OR: 3.25, 95%CI: 1.55–7.17, $p = .002$), and ≥ 2 fertility discussions during survivorship care OR: 2.71, 95%CI: 1.46–5.20, $p = .002$).

Conclusions Two-thirds of emerging adult cancer survivors expressed worry about future infertility, which has been linked to a variety of adverse quality of life outcomes. Survivors who are worried about infertility may benefit from psychological interventions.

Keywords Childhood cancer survivors · Emerging adulthood · Fertility-related distress · Fertility-related worry · Fertility status assessments · Oncofertility

There are nearly 500,000 survivors of childhood cancer living in the USA [1], many of whom may be at risk for cancer treatment-related infertility [2]. As part of their cancer treatment, children and adolescents are frequently exposed to gonadotoxic therapies (e.g., alkylating or heavy metal chemotherapy, radiation and/or surgery impacting the gonads, hematopoietic stem cell transplant), which are associated with premature ovarian insufficiency or primary ovarian failure in females and oligospermia or azoospermia in males [3, 4]. Compared with siblings, male and female cancer survivors are at increased risk for infertility [5, 6].

The risk of infertility is a source of distress for young adult cancer survivors. Uncertainty surrounding fertility and fertility-related concerns are associated with psychological distress, depression, and anxiety [7–9], particularly for younger survivors and those without children [10]. Fertility-related concerns can also negatively impact self-confidence and body image [11]. Survivors may worry that infertility will negatively impact their desirability as a partner and future romantic relationships, perceiving impending criticisms from parents and in-laws, social comparison, and societal norms surrounding biological parenthood [11, 12].

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Female cancer survivors who were diagnosed in adolescence report more fertility-related distress as compared to male survivors, including feeling upset, nervous, overwhelmed, and angry (especially in the context around uncertain fertility), having increased worries about partner reactions, and experiencing greater difficulty with daily reminders of potential infertility [13]. Survivors also express they are given little information following treatment regarding their fertility potential and fertility treatment options [14]. Furthermore, the majority of childless cancer survivors desire biological parenthood [15]. When this is not possible, some survivors experience even greater distress secondary to perceptions of health-related discrimination and/or the high financial costs associated with the adoption or surrogacy process [16].

Concerns regarding the genetics and heritability of cancer risk and fear of cancer recurrence have been reported by both male and female survivors, while females are also concerned about their ability to carry a baby to term and the potential for cardiac or other complications during pregnancy [11–13]. Cultural beliefs about parenthood and its impact on femininity and masculinity also exacerbate this distress [17]. Decreased fertility-related concerns have been reported in those who have undergone fertility preservation [13]. Unfortunately, there are significant barriers to fertility preservation for children and adolescents undergoing cancer treatment, and uptake remains limited. Fertility preservation may also not be feasible due to fears surrounding the invasiveness of the procedures, high costs, time, and delay of initiating treatment [10].

Adolescence and young adult oncology broadly encompasses the years of 15 to 39, which represent wide variation in developmental status and life experiences [18]. Emerging adulthood is a developmental period, generally between the ages of 18–25 years, representing a transition from adolescence to adulthood [19]. Emerging adults are moving toward independence and further identity formation, while experiencing milestones including completion of formal education, entering the workforce, and establishing romantic partnerships. For many survivors of childhood cancer, emerging adulthood includes a healthcare transition, from a pediatric- to adult-centered health system, along with the responsibilities of managing their own healthcare [20]. When considering the rapid and cumulative challenges of initiating and integrating multiple life goals during this sensitive period, the context is ripe for fertility-related worry and other associated concerns surrounding future family building as a cancer survivor.

To assist in this process, cancer survivors who are worried about future infertility during emerging adulthood may desire a referral for reproductive health consultation and/or psychological services while under the care of a pediatric cancer survivorship clinic. Worry about infertility among emerging adult cancer survivors has been found to be associated with interest in a fertility status assessment [21]. Identifying sociodemographic and clinical factors that are related to fertility-related worry can inform provision of psychological services and reproductive health referrals among emerging adult cancer survivors. The purpose of this study was to describe clinical and sociodemographic factors associated with expressed worry about infertility among emerging adult cancer survivors who are at risk for cancer treatment-related infertility.

Methods

As previously described [21], this study was a single-site retrospective review of clinical data from emerging adult cancer survivors (18.00–25.99 years of age) with a history of gonadotoxic treatment (alkylating agents, heavy metal chemotherapy, radiation affecting the gonads [pelvis, spine, total body irradiation, cranial radiation ≥ 30 Gy], and/or hematopoietic cell transplant). Survivors were ≥ 1 year from completion of cancer treatment and received care in the multidisciplinary survivorship clinic between 1/1/2017 and 9/30/2019. Neuro-oncology patients are followed in a separate multidisciplinary clinic and were excluded from this analysis. Ethical approval for this study was obtained from the Children's Healthcare of Atlanta institutional review board (IRB# 00,000,416) including a waiver of informed consent, as this was a retrospective chart review.

In the multidisciplinary survivorship clinic, survivors who received gonadotoxic treatment were counseled regarding their risk for infertility with a focused discussion at age ≥ 18 years, consistent with national guidelines [22, 23]. These discussions were led by the survivorship clinic's endocrinology providers and included an assessment of fertility-related worry. As a means of ecological validity, this was identified through a direct question asked during the fertility consult, "Are you worried about your future infertility?," or through expressed worry by the survivor in the consult and included in the documentation of the clinic encounter. For survivors who denied worry, this was also documented in the clinical note. These discussions also included a presentation of options for fertility status assessments, future family building, and results (normal/abnormal) of available reproductive hormones obtained as part of clinical care. Psychology support was available for survivors after the fertility discussion.

Outcomes of interest

A summary of each fertility-focused discussion was documented in the medical record using a templated clinical note. The outcome of interest for this study is worry about future infertility (yes vs. no); participants were excluded from this

analysis if documentation regarding worry was missing from their clinical note.

Sociodemographic variables

Sociodemographic variables were abstracted from participant medical records. These included sex, race/ethnicity, insurance type (managed care/commercial/private, Medicaid, Tri-care, or self-pay), religion (Christian, non-Christian), and zip code. Participants were categorized as living in a rural or non-rural area based on zip code, according to the Health Resources and Services Administration Federal Office of Rural Health Policy [24].

Clinical variables

Clinical characteristics included cancer diagnosis category (leukemia, lymphoma, solid tumor), age at diagnosis, and time from treatment completion. Treatment-related risk for infertility (low, moderate, or high risk) was determined based on each survivor's gonadotoxic exposures, using the risk stratification consensus available at the time of patient counseling [23]. Hormonal laboratory results (follicle-stimulating hormone (FSH) and anti-Mullerian hormone) were also abstracted and categorized as normal or abnormal, consistent with the interpretation and counseling provided to the survivor. Values outside of normal limits for males included a FSH $\geq$ 12.0 [25]. For females, values outside of normal limits included either an FSH $\geq$ 40.0 (indicating premature ovarian insufficiency) and/or an anti-Mullerian hormone level below the lower limit of the age- and assay-specific reference range. Receipt of a fertility consult and/or participation in fertility preservation prior to gonadotoxic treatment (yes vs. no) and the number of fertility-focused discussions (1 vs. $\geq$ 2) were also abstracted from patients' medical records.

Statistical analysis

Descriptive statistics were examined to delineate the sociodemographic and clinical characteristics of the sample, as well as to determine the proportion of participants who reported fertility-related worry. Multivariable logistic regression models were used to identify factors associated with fertility-related worry, compared with no fertility-related worry. Variables of interest (age at first fertility discussion during survivorship, age at diagnosis, time from cancer treatment completion, sex, race/ethnicity, religion, geographical location by rural status, insurance type, cancer diagnosis category, fertility preservation consult and/or procedure before treatment, hormonal laboratory evaluation, infertility risk, and number of fertility discussions during survivorship) were first examined using univariable logistic regression; variables on the univariate level with a $p < 0.10$ were considered for the multivariable model. For variables with > 5% missing data, "missing" was included as a variable category in the model. The parsimonious multivariable model was developed using stepwise backward variable elimination, only retaining variables with $p < 0.1$ in the final model. Variables in the parsimonious model with $p < 0.05$ were considered statistically significant. All analyses were conducted using R Studio version 1.3.1093.

Results

Among 308 survivors who met eligibility criteria, 49 (15.9%) were excluded due to not having completed an initial fertility-focused discussion during the study age range (18–25 years), 10 (3.3%) did not have documentation of fertility-related worry, and 249 (80.8%) were included in this analysis. There were no differences in sex or race between participants vs. non-participants (data not shown). Compared with non-participants, the participant group included a smaller proportion of leukemia survivors and a larger proportion of lymphoma survivors (36.5% of participants vs 50.8% of non-participants and 26.1% of participants vs. 11.9% of non-participants, respectively; $p = 0.037$). The final sample consisted of 139 (55.8%) male and 110 (44.2%) female participants.

At the time of first fertility discussion, participants were an average ($\pm$ standard deviation) of 19.1 ($\pm$ 1.2) years of age and 8.0 ($\pm$ 5.2) years from cancer treatment completion (Table 1). The majority of participants were non-Hispanic white (58.2%), Christian (54.6%), had commercial or private insurance (69.9%), and lived in a non-rural area (88.0%). Diagnosis categories included solid tumor (37.3%), leukemia (36.5%), and lymphoma (26.1%). Slightly more than half of participants were at low level of treatment-related infertility risk (57.0%), while 27.3% of participants were at high risk for infertility. Fifty-four (22.0%) participants had an abnormal hormonal laboratory value. Most participants (62.7%) completed one fertility-focused discussion during a survivorship clinic visit (Table 1).

The majority of participants (66.3%) reported worry about infertility. In the parsimonious multivariable logistic regression model, factors associated with expressed worry about future infertility included female sex (odds ratio [OR] 2.64, 95% confidence interval [CI]: 1.44–4.96, $p = 0.002$), a diagnosis of solid tumor (OR = 2.31, 95%CI: 1.15–4.71, $p = 0.019$), moderate level (OR = 2.94, 95%CI: 1.23–7.64, $p = 0.020$) or high level (OR = 3.25, 95%CI: 1.55–7.17, $p = 0.002$) of risk for infertility, and $\geq$ 2 fertility discussions during survivorship (OR = 2.71, 95%CI: 1.46–5.20, $p = 0.002$; Table 2). The variables time from cancer treatment completion and hormonal laboratory evaluation met

Table 1 Participant characteristics ($N = 249$)

Characteristic	Entire sample ($N = 249$)	
Age at first fertility discussion during survivorship (years)		
Mean (SD)	19.1	1.2
Median (range)	18.7	18.0, 25.3
Age at diagnosis (years)		
Mean (SD)	9.6	5.6
Median (range)	10.3	0.1, 21.9
Time from cancer treatment completion (years)		
Mean (SD)	8.0	5.2
Median (range)	6.7	1.0, 21.5
	N	%
Sex		
Male	139	55.8
Female	110	44.2
Race/ethnicity		
Non-Hispanic white	145	58.2
Black	64	25.7
Hispanic	21	8.4
Asian	8	3.2
Other	11	4.4
Religion		
Christian	136	54.6
Non-Christian or no preference	60	24.1
Missing	53	21.3
Geographical location by rural status		
Non-rural	219	88.0
Rural	30	12.0
Insurance type		
Commercial/private	174	69.9
Medicaid/self-pay	75	30.1
Cancer diagnosis category		
Leukemia	91	36.5
Lymphoma	65	26.1
Solid tumor	93	37.3
Fertility preservation consult and/or procedure before treatment		
Yes	28	11.2
No	221	88.8
Hormonal laboratory evaluation[†]		
Within normal limits	192	78.0
Outside normal limits	54	22.0
Infertility risk		
Low	142	57.0
Moderate	39	15.7
High	68	27.3
Number of fertility discussions during survivorship		
1	156	62.7
≥ 2	93	37.3

SD = standard deviation

[†]Proportions were calculated by excluding missing values ($n = 4$). Hormonal laboratory values outside of normal limits for males included a follicle-stimulating hormone (FSH) ≥ 12.0 and for females included either an FSH ≥ 40.0 and/or an anti-Mullerian hormone level below the lower limit of the assay-specific reference range

[±]Survivors were asked if they were worried about future infertility as part of the fertility discussion

Table 2 Multivariable parsimonious model of factors related to expressed fertility-related worry

Variable	OR	95% CI		*p* value
Sex				
Male	1.00			
Female	**2.64**	**1.44**	**4.96**	**.002**
Cancer diagnosis				
Leukemia	1.00			
Lymphoma	1.28	0.62	2.65	.506
Solid tumor	**2.31**	**1.15**	**4.71**	**.019**
Infertility-risk level				
Low	1.00			
Moderate	**2.94**	**1.23**	**7.64**	**.020**
High	**3.25**	**1.55**	**7.17**	**.002**
Number of survivorship fertility discussions				
1	1.00			
≥2	**2.71**	**1.46**	**5.20**	**.002**

OR, odds ratio; *CI*, confidence interval

Using backwards elimination, time from cancer treatment completion and hormonal laboratory evaluation were eliminated from the parsimonious model. Variables in the parsimonious model that met statistical signficance (p < .05) are bolded

criteria for entry into the multivariable model but were eliminated using backwards stepwise elimination during the development of the parsimonious model. Worry about infertility was reported by 75.5% of females, 78.5% of survivors of solid tumors, and 76.9% and 80.9% of survivors at moderate and high risk for treatment-related infertility, respectively, and 79.6% of survivors who had ≥ 2 fertility-focused discussions during survivorship (Fig. 1).

Discussion

In this study of emerging adult cancer survivors who received a fertility-focused discussion during a pediatric cancer survivorship clinical visit, nearly two-thirds reported fertility-related worry. Factors associated with worry were female sex, a cancer diagnosis of solid tumor, moderate and high risk of infertility, and having received two or more fertility-focused discussions during survivorship care.

Consistent with prior research in cancer survivors, we found that females were more likely to report fertility-related worry as compared to males. This was true across all diagnoses, risk groups, and number of discussions. In a study by Gilleland Marchak et al., the majority of female cancer survivors (71%) and their parents were worried about infertility following gonadotoxic therapies [26]. Female survivors of childhood and adult cancers have been identified as having more information needs about fertility-related topics,

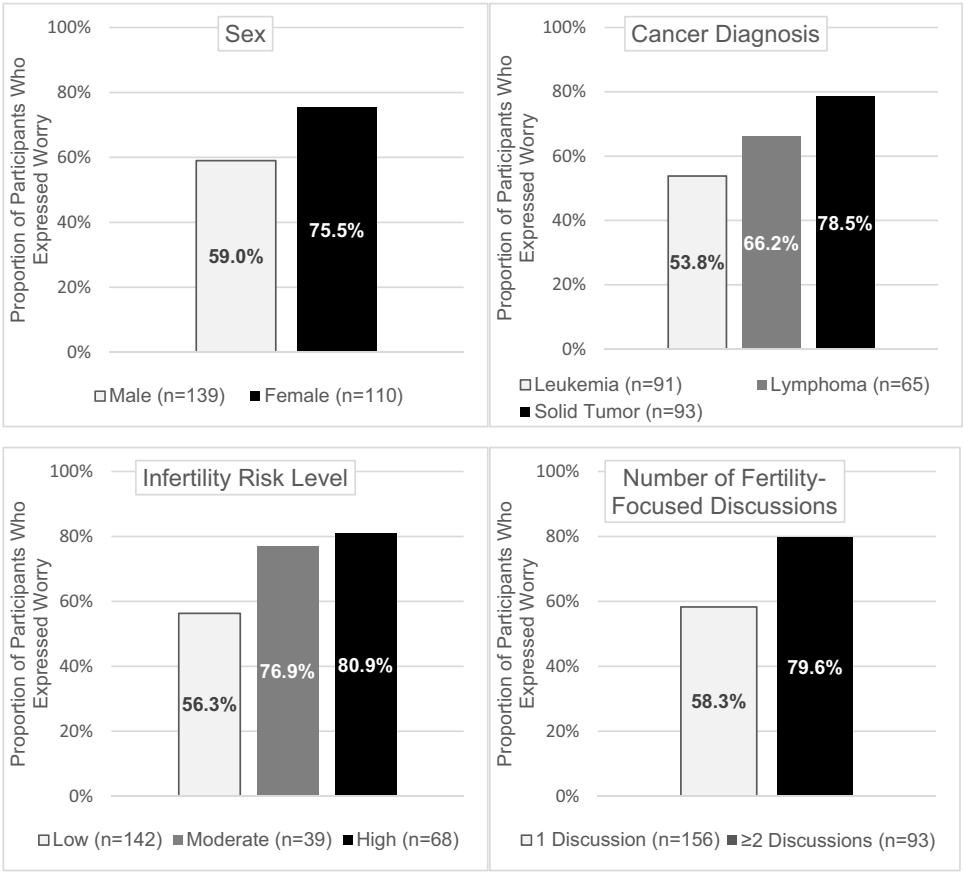

Fig. 1 Proportion of participants who expressed fertility-related worry by sex, cancer diagnosis, infertility-risk level, and number of fertility-focused discussions

which along with increased reproductive concerns have been linked to lower quality of life [27, 28]. Female sex has also been associated with adverse psychological symptoms in survivorship [29]. Furthermore, female survivors have expressed that they would experience increased distress if parenting goals remained unfulfilled [30]. This information demonstrates the importance of meeting educational needs in survivorship care, psychological support for female cancer survivors surrounding risk for infertility, and a discussion of family building options.

Cancer treatment factors that were associated with worry included higher risk for treatment-related infertility (moderate and high risk) or a diagnosis of solid tumor. These factors are related—survivors were classified into risk categories based on the treatment they received which is dependent on their diagnosis, age, and response to therapy. In this sample, it is difficult to tease these factors apart given the heterogeneity in the broad diagnosis categories and their treatment. Compared with survivors of leukemia, survivors of solid tumors were more likely to report worry about infertility. Solid tumors represent a broad diagnosis category, including sarcomas, which are treated with multiple modalities (chemotherapy, surgery, radiation) according to diagnosis, stage, location of disease, and treatment protocol. Although the solid tumor group is made up of a heterogeneous group of tumors, the statistically significant association with fertility-related worry remained when controlling for infertility-risk category.

Counseling regarding risk for infertility is recommended to occur at the time of cancer diagnosis and through survivorship [31]. Stratification of infertility-risk is based on therapeutic exposures and is recommended to be incorporated within counseling [32]. Our research team has demonstrated that individualized education during fertility-focused discussions in survivorship improves survivors' knowledge of their treatment-related infertility-risk level [23]. Therefore, it is not surprising that survivors who are at moderate or high risk for infertility are more likely to report worry about infertility. It is also important to note that about 20% of survivors in our sample who are at moderate and high risk for infertility did not report worry, while half of those at low risk did report worry. This suggests that while stratified risk is an important factor related to worry, there is variation among survivors regarding worry across treatment exposure groups. Survivors without exposure to gonadotoxic therapies were excluded from our study, although worry about infertility can extend to survivors who are not at risk for treatment-related infertility [33].

Survivors who received more discussions about fertility were also more likely to experience worry. As they age, emerging adults may have increased concerns about the ability for biological parenthood. Furthermore, during emerging adulthood, opportunities arise for additional conversations focused on their infertility risk, with conversations now directed to them instead of their parents. Survivors awareness of infertility risk is associated with worry [26], therefore report of worry by the survivor may signify an acknowledgement of their risk for infertility and reflect their prioritization of biological parenthood. It is possible that survivors who are worried about infertility were more likely to participate in repeated fertility-related discussions, while for some survivors, repeated discussions may have resulted in worry. The temporal relationship between timing of fertility-related discussion and worry, and the prevalence of distress associated with worry, should be investigated in prospective research. In the clinical setting, if significant risks for infertility are identified and survivors have strong parenthood desires, conversations about alternative methods of family building, including adoption or surrogacy, are warranted. Psychologists, as members of the multidisciplinary survivorship clinic, can work with survivors to address reproductive distress.

Worry about infertility has been shown to be associated with interest in a fertility status assessment among emerging adult cancer survivors [21], suggesting that survivors who express worry may benefit from referral to a reproductive specialist for an assessment. However, completion rates of fertility status assessment are low among survivors which warrants further study to identify barriers and decisional factors that influence assessment completion [21]. Survivors may fear the findings of the fertility status assessment and avoidance of potential negative findings may result in continued uncertainty, further worsening worry [30, 34]. On the other hand, worry may be mitigated for survivors who pursue an assessment and receive findings that are encouraging regarding future fertility potential. Psychological support is recommended following fertility status assessment, regardless of findings, to help process potentially difficult news and cope with planning for next steps and options for family building.

Worry may also impact engagement in other aspects of their survivorship care, such as regular survivorship visit attendance, following through with needed medical surveillance and referral to specialists, and active participation in their care. Due to the strong relationship between physical and psychological health, distress in survivorship may be related to poorer psychological functioning, increased medical utilization, and maladaptive health behaviors [35]. Identifying and providing support for worried survivors through regular screening could help to increase survivor engagement during emerging adulthood as they work to transition to adult survivorship programs where they will continue to take more responsibility for their care. For some survivors, repeated discussions regarding potential infertility, in combination with supportive psychological interventions, could potentially assist with reduction of fertility-related worry. If

infertility concerns are identified, then connecting survivors with adult reproductive endocrinology providers and psychological support services is warranted.

Clinical implications

These findings indicate that a majority of childhood cancer survivors are worried about their future fertility during the emerging adulthood period, a developmental stage encompassing the transition from adolescence to adulthood and the transition to adult healthcare systems. Even though these survivors are newly entering adulthood, the majority is reporting worry about infertility, despite being potentially years from initiating family building. Male and female survivors who were worried about future infertility had a 3- and 4- fold increased likelihood of expressing interest in a fertility status assessment respectively [21]. As survivors begin to seek and form committed romantic relationships and navigate the expectations of others regarding biological parenthood, emerging adulthood may be a time when they are interested in receiving further information about their fertility status through a formal assessment to help potentially mitigate reported worry [36]. Providing an opportunity to complete a fertility status assessment during emerging adulthood, prior to transitioning to adult healthcare systems, may help to reduce uncertainty and distress among survivors as they age. These findings also support the need for psychological intervention and future planning to discuss options for family building.

Limitations

There are limitations in this study that are important to note and address in future research. First, the worry measurement was abstracted from clinic documentation and identified participants as either having expressed worry or not expressing worry about fertility. Although this is an ecologically valid approach to studying worry, future research would benefit from a quantitative metric of worry to further explore the intensity of reported worry and its relationship to predictors. Participants who were missing worry documentation from their clinical note ($n = 10$) were unable to be included in the study. Secondly, the study was a retrospective single-site design that did not include brain tumor survivors, who were followed in a separate survivorship clinic. The sample for this study includes only patients who attended our institution's survivor clinic. While 70% of eligible patients attend our survivor clinic [37] and nearly 90% of our established survivors returned for annual survivorship care [38], this sample does not represent those who do not attend or who become lost to follow-up prior to age 18 years. Expansion of this research in multiple sites and with other survivorship populations would allow for more in-depth examinations of predictors of fertility-related worry.

Conclusions

With the number of survivors of childhood cancer increasing, it is important to meet the needs of survivors in relevant areas, including potential infertility. With worry about infertility being a relevant factor to many survivors, and particularly prevalent in females, emotional support by psychological professionals is needed. Screening for distress among survivors who are worried about future infertility may be beneficial to identify individuals in need of further support and connect them with appropriate psychological services. Collaboration between the medical team and mental health professionals is necessary to provide optimal, multidisciplinary care to support survivors of childhood cancer during their emerging adulthood years.

Author contribution All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Brooke Cherven, Erin Kelling, Rebecca Lewis, and James Klosky. The first draft of the manuscript was written by Brooke Cherven and Erin Kelling, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval This study received ethical approval from the Children's Healthcare of Atlanta institutional review board (IRB# 00000416) and was conducted in accordance with the Declaration of Helsinki.

Competing interests All authors report no competing interests.

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